

Precedence — Flight ladder (Eras 11–20) curriculum draft

Math Sensei · 3 Oct 2026 · design only — not built, not on Pages yet

Premise. Same game loop as Precedence: the player never does the arithmetic. They only tap **what comes next** in a brutally nested expression. A calculator (or the collapse engine) does the “how much.” This ladder trains the **other** brain region — the one that fails on hard precalc and first-year calc exams: **seeing structure under pressure**.

Band. Eras 11–20 after the ground/road ten (Sums → Fraction bars). Miami Burnout (Era 8) stays the road **1.0**. Flight is the **1.5 / advanced** wing — unlocked after Fraction bars (or after a Road Test on Era 10).

UI titles stay math. Flight craft are **visual-only** (same lock as Cleopatra litter). Garage cards read the math titles below; the hangar silhouette tips the fantasy.

Nomenclature (unchanged). Era → Levels (stages) → Run → Step. Algebra Licenses remain a separate wing.

1. Ten flight metaphors (historical progression)

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Era · Craft (visual only) · Place / vibe · Fantasy beat
---: · --- · --- · ---
11 · Wright Flyer · Kitty Hawk dunes, 1903 · First leave the ground — still fragile
12 · Barnstormer biplane · County fair, 1920s · Showmanship, loops, low passes
13 · Flying boat / Clipper · Harbor at dusk, 1930s · Long-haul romance, map rooms
14 · 1950s prop airliner · Cabin with turkey dinner cart · Nostalgic cruise altitude — classy, not frantic
15 · Early jet airliner · 1958–60s silver tube · Smooth, fast, glassless cockpit
16 · Supersonic airliner · Concorde needle, Mach haze · Compressed time — denser expressions
17 · Modern glass-cockpit airliner · Night approach, PAPI lights · Instruments everywhere; read the panel
18 · Afterburner fighter · High-G blue · Aggressive chain rules; speed as streak
19 · Stealth bomber · Radar-dark silhouette · Hidden structure; what you *don't* see first
20 · Autonomous drone / swarm · HUD reticles, no pilot · Elite: many moving parts, one legal next step
```

(Orville Redenbacher stays in the snack cart on Era 14, not on the fuselage.)

2. Curriculum map — math titles + skills

A. Precalc order-mastery (Eras 11–15)

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Era · Math title · What “next” means · Classic exam fails it trains · Emphasized skills (Mathera-ish)
---: · --- · --- · --- · ---
**11** · Grouping towers · Innermost grouping first across `( )`, ` ` , ` ` , ` ` , and nested radicals ·
Treating  $\sqrt{a+b}$  like  $\sqrt{a}+\sqrt{b}$ ; abs vs parens; “outside first” · Nested grouping; absolute value as grouping;
radical as unary wrapper — extend `III.1.08` / precalc grouping IDs
**12** · Fraction architecture · Which bar / vinculum binds first; complex fractions as towers · Cancelling
across addition; attacking the wrong floor of a complex fraction · Complex fractions; rational nesting; extend
Era 10 fraction-bar line
**13** · Composition order · Read ` $f(g(h(x)))$ ` and ` $(f \circ g)(x)$ ` left/right correctly; inverse as undo order
· Computing  $g(f(x))$  when asked  $f(g(x))$ ; inverse steps backwards · Function composition; inverses as reverse
order — new composition skill IDs
**14** · Trig reading · ` $\sin^2 x$ ` vs ` $\sin(x^2)$ ` vs ` $\sin^{\square} x$ `; degree/radian is *not* the point — power vs
composition is · Squaring the wrong thing; inverse trig vs reciprocal · Trig notation literacy (power,
composition, inverse) — testing-mode trig grain
**15** · Exp-log stacks · Log of a power vs power of a log; change-of-base as regrouping · Moving exponents
through logs illegally; expanding too early · Exp/log laws as *order* moves; change of base as rewrite sequence
```

Hard-prec calc exam spirit: IB AA HL / AP Precalc / UK A-level pure “simplify / rearrange without evaluating” items.

B. Calculus class (Eras 16–20)

User ask: limits, derivatives, integrals + **two more**. Proposal for the pair: **infinite series structure** and **multivariable / implicit chain** (the two places order errors stay lethal after Calc I).

```
Era · Math title · What “next” means · Classic fail · Why this pair fits
---: · --- · --- · --- · ---
**16** · Limits structure · Algebraic legal next step *inside* a limit (factor → cancel → plug;
conjugate; dominant term) before “evaluate” · Plugging first; canceling terms that aren’t factors · Glass
cockpit: read instruments before trusting the number
**17** · Derivative protocol · Which differentiation rule applies *next* (const, sum, product, quotient,
chain) — pick the rule, don’t compute the derivative by hand · Chain last when it should be first; quotient vs
```

product misread · Afterburner: one wrong rule order = flameout
****18**** · ****Integral protocol**** · Recognition order: rewrite → choose technique (u-sub vs parts vs partial fractions) → execute skeleton · Parts when u-sub; differential in the wrong place · Stealth: the technique is half-hidden until you see structure
****19**** · ****Series structure**** · Term-by-term legality; geometric vs Taylor skeleton; what you may differentiate/integrate termwise · Rearranging divergent series; wrong center; factorial vs exponential growth misread · Drone swarm: many terms, one legal move
****20**** · ****Several variables**** · Partial vs total; which variable is held fixed; implicit / related-rates dependency order; mixed partials equality as a *check*, not a first move · Treating ∂ as d ; chain rule missing a path · Elite hangar: no single “forward” — you choose the axis

Alternate for Era 19–20 if you want to stay single-variable longer: **19 Related rates / implicit, 20 Diff EQ / separation order**. I'd still prefer series + multivariable for “hard exam that still isn't about punching the calculator.”

3. Stages within each Era

Recommendation: 6 stages per Era (plus an optional 7th mixed road test).

Stage # · Role · Purpose
 ---: · --- · ---
 1 · ****Lexicon**** · One new notation habit in isolation (e.g. only $\sin^2$ vs $\sin()$)
 2 · ****Trap set A**** · Classic wrong-first moves as distractor cards / wrong taps
 3 · ****Trap set B**** · Second failure mode (different shape, same skill)
 4 · ****Combine**** · New habit + previous Era's habit in one expression
 5 · ****Dense**** · Exam-length nests; still no arithmetic grind
 6 · ****Fluency**** · Timed Rally / Forks; streak = altitude
 7 (opt.) · ****Road test / Checkride**** · 12 mixed; 1 slip OK → stamp Era license / unlock next craft

Why 6 not 10–12: Algebra Licenses already run ~10–15 levels deep for *rewrite* pedagogy. Flight eras are **protocol trainers** — fewer stages, nastier trees, more repetition via generator tiers (0–2) rather than more named stages. $6 \times 10 = 60$ **stages** for the whole flight wing; with tiers that's plenty of runway without feeling like a second ALEKS.

Stamps: keep Practice-can-clear (e.g. 8 clean → stamp, 4 stamps → Era cleared) — slightly tighter than road eras because each stage is denser.

4. What stays out of scope (on purpose)

- Numeric evaluation, CAS grinding, GDC button sequences (that's Testing Mode / Notebooks).
- Full proof writing / FRQ paragraphs (overlay later).
- Teaching *new* calculus ideas from zero — Flight assumes the idea exists; it trains **order under complexity**.
- Replacing Algebra Licenses — those stay the rewrite/equivalence wing.

5. Suggested unlock fantasy

- Eras 1–10: ground travel (visual).
- Clear Era 10 (or its checkride) → **hangar opens**.
- Eras 11–15: props → early jet (precalc).
- Eras 16–20: glass → drone (calc).
- Era 20 elite tag; optional Pro ★ checkride = “Airline Transport” joke badge.

6. Open locks for Julius

1. Confirm Era 19–20 = **Series + Several variables**, or swap to Implicit/Related rates + Diff EQ.
2. Confirm **6 stages + optional checkride** per Era.
3. Flight wing name: **Flight, Hangar, Altitude**, or still just Eras 11–20 under Precedence?
4. Unlock gate: after Fraction bars only, or also require Miami (Era 8) ★?

7. One-page skill spine (for later tagging)

Precalc five: Grouping towers · Fraction architecture · Composition order · Trig reading · Exp–log stacks

Calc five: Limits structure · Derivative protocol · Integral protocol · Series structure · Several variables

Next step when you say go: freeze this into ERAS stubs 11–20 (locked cards) on the site — still no generators until content pass.